Ishan Gupta

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Mechanical Engineering — Dec 2026
Courses: Modern Control Theory, ML/AI for Engineers, Intro to Robot Learning — GPA: 3.76/4.00

The University of Texas at Austin — Austin, TX
Bachelors of Science in Mechanical Engineering | Robotics Minor | Certificate in Computer Science — May 2025
Courses: Systems and Controls, Mechatronics, ML for Engineers, Intro to Robotics — GPA: 3.88/4.00

SKILLS

Programming: Python, Matlab, Simulink, C++, Scipy, PyTorch, Sklearn, PyBullet, Gymnasium
Software Tools: ROS2, CARLA, Mujoco, Webots, Solidworks, Arduino, Excel, Powerpoint, Blender

ACADEMIC PROJECTS

F1Tenth Autonomous Racing / Pittsburgh, PA — Jan 2026 - Present
- Developed ROS2-based perception and autonomy modules in C++ and python for an F1TENTH vehicle, integrating LiDAR processing, localization, and closed-loop control for indoor autonomous racing.
- Implemented LiDAR-based SLAM/localization to support GPS-denied navigation, enabling map-based autonomous driving and real-time pose estimation on indoor tracks.
- Trained a YOLO detector for “car-behind-ego” detection using a custom dataset collected from F1TENTH runs and validated using mAP/precision-recall with iteration on augmentation for generalization.

Learning-Based Behavior Planning for Urban Driving / Pittsburgh, PA — Jan 2026 - Present
- Developed a PPO behavior planner in CARLA and evaluated robustness using success/collision metrics across varying traffic densities; automated rollout logging and analysis to compare policy iterations.
- Designed observation and reward formulations to reason over ego state, nearby traffic, safety margins, and uncertain interaction outcomes, while interfacing the learned planner with low-level trajectory control.
- Achieved 3% higher success rate with PPO local planner compared to baseline for highway merging scenario.

Simulated Autonomous Vehicle Control / Pittsburgh, PA — Oct 2025 - Dec 2025
- Developed PID and pole placement algorithm to autonomously control a car around a track with an average of 0.47 meters discrepancy from the trajectory in Webots.
- Improved controls using LQR algorithm and A star to remapping around a car, reducing time of completion by 20%.
- Implemented EKF-based state estimation / SLAM for autonomous navigation, enabling localization and trajectory tracking with average path deviation of 0.7 m.
- Designed an MRAC with a baseline LQR controller to fly a quadrotor with up to 68% loss of thrust in one motor.

PROFESSIONAL EXPERIENCE

Test Engineer Intern / The Orbex Group / Austin, TX — May 2025 - Aug 2025
- Built and debugged back-to-back motor test rig for loaded motor evaluation, assembling frames, wiring motors and torque sensors, and reducing vibration through iterative mechanical troubleshooting.
- Analyzed motor, vibration, and environmental test data in Excel, and created macro-based plotting tools to automate future test analysis workflows.
- Supported root-cause analysis on nonconforming motors and test rigs, including coupler alignment, vibration issues, encoder/current-spike troubleshooting, and process-improvement observations.

Robotics Engineer Intern / Pike Robotics / Austin, TX — Feb 2023 - Aug 2024
- Secured a $120K grant by leading proposal materials including BOM, conceptual CAD, and Blender animation to communicate the robot concept to stakeholders.
- Designed a mechanical multiplexer ejection system and iterated sensor-mount designs, improving system scalability, sensor coverage, and reliability.
- Conceptualized a two-robot tether/anchor system and developed an actuated wedge mechanism that reduced inspection setup time by 10%.